But with the enzyme catalyst the activation energy of this reaction is less than 2kcal/mole.

(ii) Enzyme catalysis is highly specific in nature.

$H_2N\text{-}CO\text{-}NH_2 + H_2O \rightarrow 2NH_3 + CO_2$

The enzyme urease which catalyses the reaction of urea does not catalyse the following reaction of methyl urea

$H_2N\text{-}CO\text{-}NH\text{-}CH_3 + H_2O \rightarrow$ No reaction

(3) Enzyme catalysed reaction has maximum rate at optimum temperature. At first rate of reaction increases with the increase of temperature, but above a particular temperature the activity of enzyme is destroyed. The temperature at which enzymic activity is high or maximum is called as optimum temperature.

**For example:**

- Enzymes involved in human body have an optimum temperature 37°C /98°F
- During high fever, as body temperature rises the enzymatic activity may collapse and lead to danger.

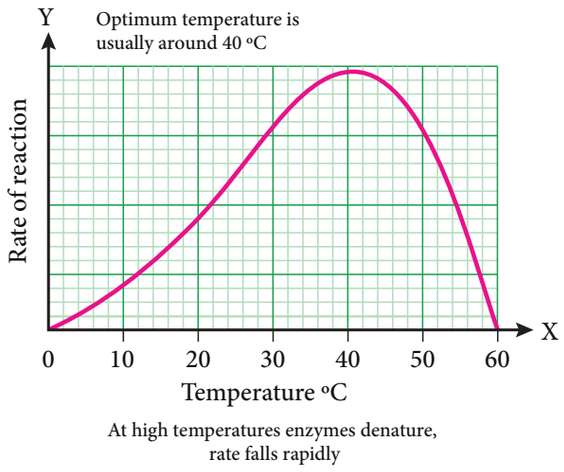



**Figure 10.6 Rate vs Temperature**

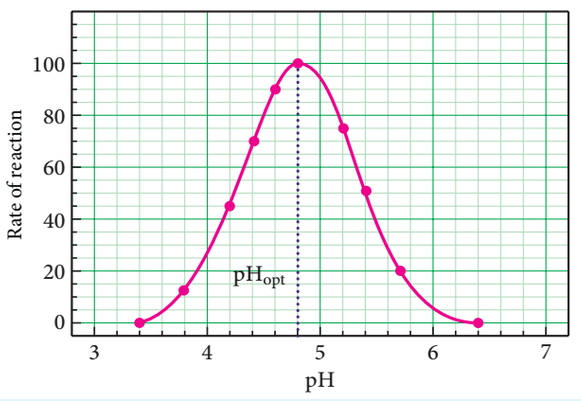



**Figure 10.7 Rate vs pH**

4. The rate of enzyme catalysed reactions varies with the pH of the system. The rate is maximum at a pH called optimum pH.
5. Enzymes can be inhibited i.e. poisoned.

Activity of an enzyme is decreased and destroyed by a poison.

The physiological action of drugs is related to their inhibiting action.

Example: Sulpha drugs. Penicillin inhibits the action of bacteria and used for curing diseases like pneumonia, dysentery, cholera and other infectious diseases.

6. Catalytic activity of enzymes is increased by coenzymes or activators.

A small non protein (vitamin) called a coenzyme promotes the catalytic activity of enzyme.

## 10.4 Zeolite Catalysis:

The details of heterogeneous catalysis will be incomplete, if zeolites are not discussed. Zeolites are microporous, crystalline, hydrated, alumino silicates, made of silicon and aluminium tetrahedron. There are about 50 natural zeolites and 150 synthetic zeolites. As



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